

Assignment-3

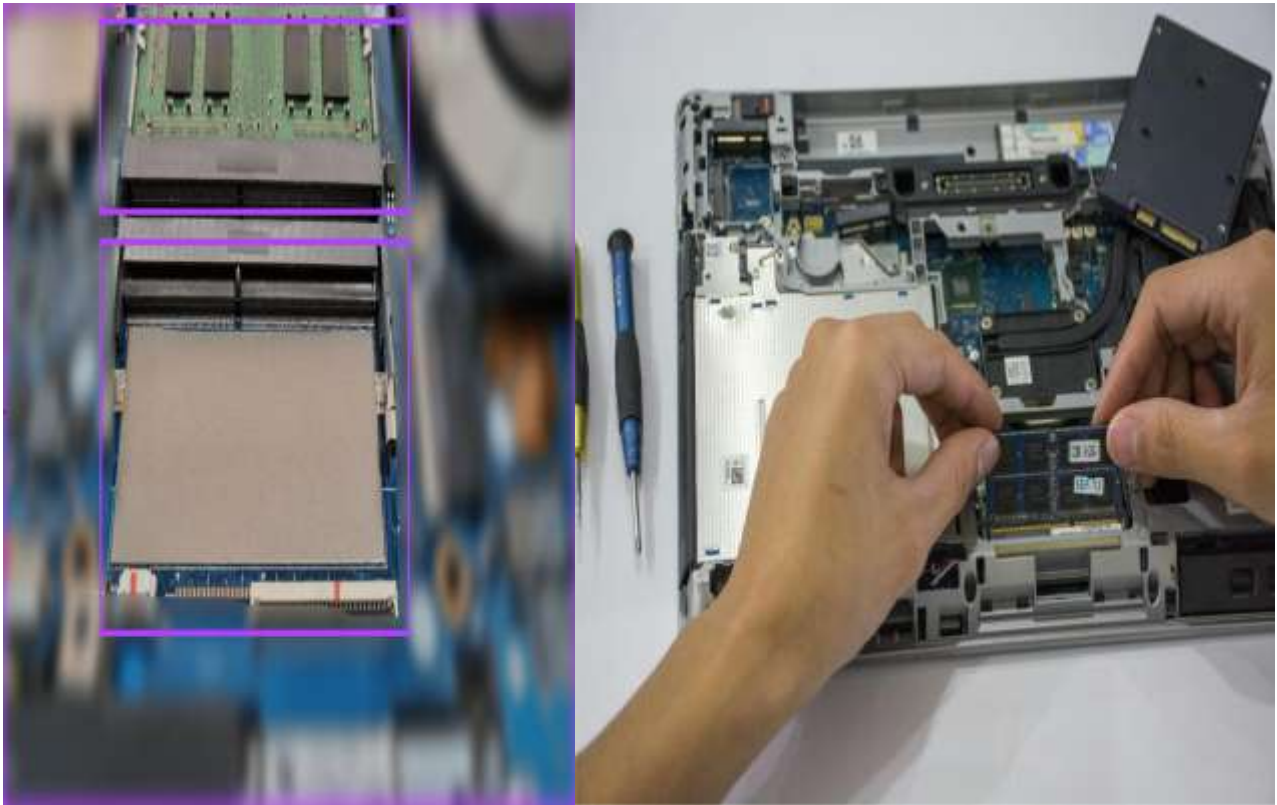
1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Simple Analogy	An older two-lane road	A modern four-lane highway	A super-fast bullet train
Speed (Transfer Rate)	Slowest (800 – 2,133 MT/s)	Fast (2,133 – 3,200+ MT/s)	Ultra Fast (4,800 – 8,400+ MT/s)
Voltage (Power Used)	Highest: 1.5 Volts <i>(Uses the most power & gets warmer)</i>	Lower: 1.2 Volts <i>(Saves energy and runs cooler)</i>	Lowest: 1.1 Volts <i>(Most power-efficient)</i>
Max Capacity (Per Stick)	Up to 8 GB – 16 GB	Up to 32 GB – 64 GB	Up to 128 GB+
Typical Use Cases	Old computers, basic office PCs, legacy laptops	Standard modern laptops, mid-range gaming PCs, daily work	High-end gaming PCs, 4K video editing, 3D design, AI workloads

Key Points to Remember

- **Compatibility:** They are not interchangeable. Each type has its notch in a different physical spot on the stick, so a DDR4 stick will not fit into a DDR3 or DDR5 motherboard slot.
- **Speed:** DDR5 is roughly twice as fast as standard DDR4.
- **Power Efficiency:** Higher generations use less power while delivering much faster speeds.

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully. Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



❖ Steps to Remove and Reinstall Laptop RAM

1. Shut down the laptop completely.
2. Disconnect the charger and any other cables.
3. Open the back cover of the laptop carefully.
4. Locate the RAM slot on the motherboard.
5. Push the two side clips outward to remove the RAM module.
6. Align the notch of the RAM with the RAM slot.
7. Insert the RAM at a slight angle (about 30–45°).
8. Press the RAM down until both side clips lock into place.

- 9. Close the back cover and tighten the screws.
- 10. Turn on the laptop and check that it boots successfully.

Verification

- ✓ The RAM was reinstalled correctly.
- ✓ The laptop powered on successfully.
- ✓ The operating system loaded normally without any errors.

3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Differences Between HDD, SSD, and NVMe

Feature	HDD	SSD	NVMe SSD
Speed	Slow	Fast	Very Fast
Technology	Spinning magnetic disk	Flash memory	Flash memory with PCIe interface
Best Use	Storing large files	Everyday computing	Gaming, video editing, heavy applications

Daily App / Use Case

Storage Type	Example App	Why It Benefits
HDD	Movies, Photos, Music Storage	Good for storing large files at a low cost.
SSD	Google Chrome	Opens quickly and makes web browsing faster.
NVMe SSD	Adobe Photoshop	Loads large files very quickly and improves editing performance.

➤ Short Conclusion

- HDD is best for storing large amounts of data.
- SSD is best for fast everyday use and quick startup.
- NVMe SSD is best for high-performance tasks like gaming, video editing, and professional software.

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly. Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

LAPTOP SSD REPLACEMENT – STEP BY STEP

1 Turn off the laptop and remove the back cover. 	2 Open the back cover. 	3 Locate the storage drive. 
4 Disconnect the storage drive. 	5 Remove the storage drive. 	6 Identify the connector (SATA). 
7 Install the new storage drive. 	8 Reconnect and close the back cover. 	9 Power on and verify the SSD. 

✓ The new SSD is installed and detected successfully.

Steps Performed

1. Shut down the computer.
2. Disconnect the power cable.
3. Open the laptop/PC case.
4. Locate the storage drive (HDD/SSD).
5. Identify the connector type (SATA or M.2/NVMe).
6. Carefully remove the storage drive.
7. Take a clear photo of the connector.
8. Reinstall the storage drive securely.
9. Close the case and reconnect the power.
10. Turn on the computer and verify that the storage device is detected.

Verification

- ✓ The storage drive was reinstalled successfully.
- ✓ The device booted normally.
- ✓ The operating system detected the storage drive without any errors.